

I B.Tech I Semester Regular & Supplementary Examinations, January - 2025
Basic Electrical and Electronics Engineering
(Common to EEE, ECE, ECE(BM), IT, CSE(AI), CSE(CS), CSE(IOT) & CSE(BS))
(For 2023 & 2024 Admitted Batches Only)

Time: 3 Hours

Max.Marks:70

PART-A Basic Electrical Engineering**SECTION-I****Answer all questions and all questions carry equal marks****5 x 1 = 05M**

- 1.a) Define Ohm's Law.
- 1.b) Define Power factor.
- 1.c) Write the principle of generator.
- 1.d) Mention the advantages of MI Instruments.
- 1.e) Define Fuse.

SECTION-II**Answer All questions and all questions carry equal marks****3 x 10 = 30M**

- 2 a) Derive an expression for total resistance when three resistances R₁, R₂ & R₃ are connected in Series?
(OR)
- b) i. State and explain Superposition theorem.
ii. Derive the expressions for form factor.
- 3 a) Explain the construction and working of a three-phase induction motor with a neat sketch and its applications.
(OR)
- b) Derive the expression for balance condition of Wheatstone bridge.
- 4 a) Draw and explain the block diagram of solar and wind power plant.
(OR)
- b) i. Define tariff? Explain the two-part tariff with an example.
ii. Explain MCB and its applications.

PART-B Basic Electronics Engineering**SECTION-I****Answer all questions and all questions carry equal marks****5 x 1 = 05M**

- 5.a) What is the important application of a Zener Diode.
- 5.b) Draw the symbols of PNP and NPN transistors.
- 5.c) What is the advantage of Bridge rectifier over a center tapped full wave rectifier
- 5.d) What is the use of a Loud speaker.
- 5.e) What is a combinational circuit

SECTION-II**Answer All questions and all questions carry equal marks****3 x 10 = 30M**

- 6 a) What is a diode. Explain voltage current characteristics of a PN junction diode with suitable diagrams
(OR)
- b) Draw and explain the input and out characteristics of a BJT in common Base Configuration with the help of circuit diagram
- 7 a) What is Zener effect. Explain how the Zener diode can be used as a voltage regulator with necessary circuit diagram
(OR)
- b) Explain the block diagram of electronic instrumentation system
- 8 a) Draw the symbols of OR, AND, NOT, NOR, NAND, XOR and XNOR gates. Write their logic expressions and truth tables
(OR)
- b) Draw the circuit diagram of full adder with two adders and explain its operation with truth table.